Pré-processamento dos dados

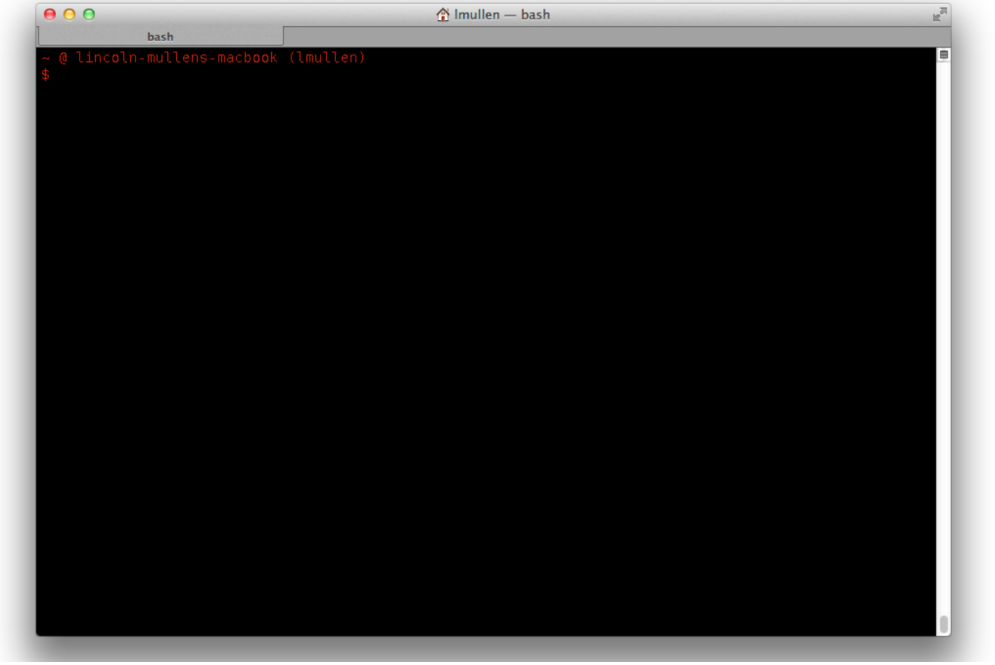


Pré-processamento dos dados : Generalidades

Programas gratuitos em linha de comando

Geralmente desenvolvidos por biólogos/ bioinformatas/instituições acadêmicas

Fácil integração com computação de alta performance (HPC)



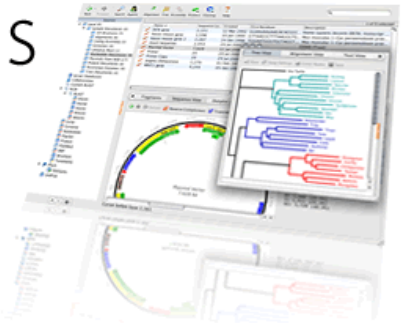
Pré-processamento dos dados : Generalidades

Alternativas pagas com interface gráfica

Agregam programas de uso livre

Capacidade computacional limitada

geneious

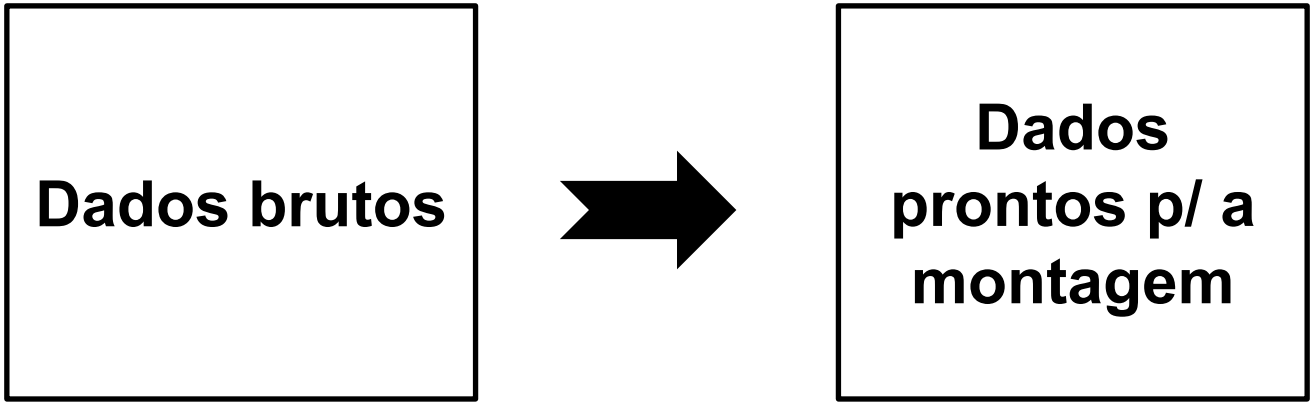


Personal Subscriptions

1 year subscription plans for individual use.

STUDENT PERSONAL	ACADEMIC PERSONAL	CORPORATE PERSONAL
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Pré-processamento dos dados : Generalidades



Poda (trimming) de qualidade
Poda (trimming) dos adaptadores e índices
Parear os reads forward and reverse
Colapsar leituras redundantes
Eliminação dos contaminantes
Eliminação dos elementos repetitivos

Pré-processamento dos dados : Análise de qualidade

Manipular arquivos FASTQ

FastQC

http://www.bioinformatics.bbsrc.ac.uk/projects/fastqc/

Análise e controle de qualidade

Pré-processamento dos dados : Análise de qualidade

Manipular arquivos FASTQ

FastQC Report

Summary

- Basic Statistics
- Per base sequence quality
- Per sequence quality scores
- Per base sequence content
- Per base GC content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Kmer Content

Basic Statistics

Measure	Value
Filename	bad_sequence.txt
File type	Conventional base calls
Encoding	Illumina 1.5
Total Sequences	395288
Sequence length	40
%GC	47

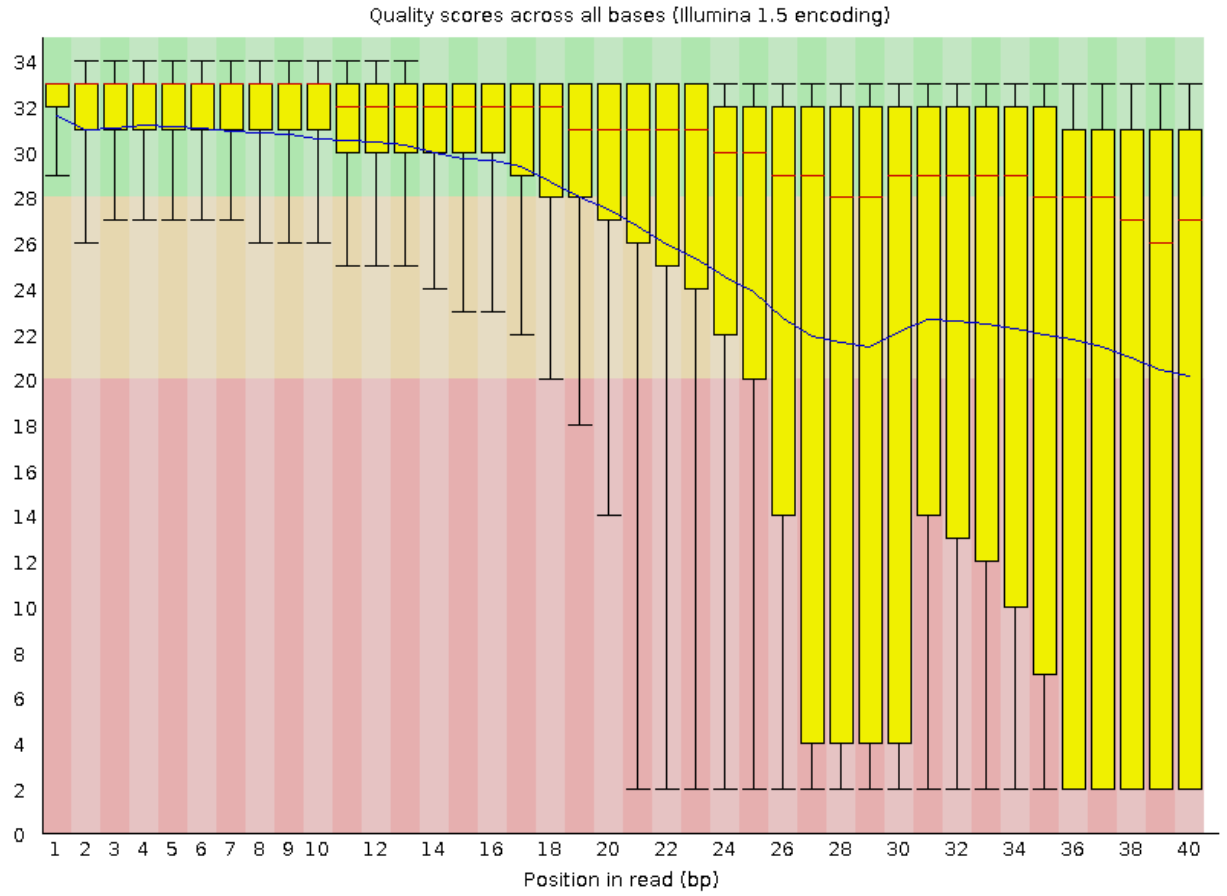
Pré-processamento dos dados : Análise de qualidade

Manipular arquivos FASTQ

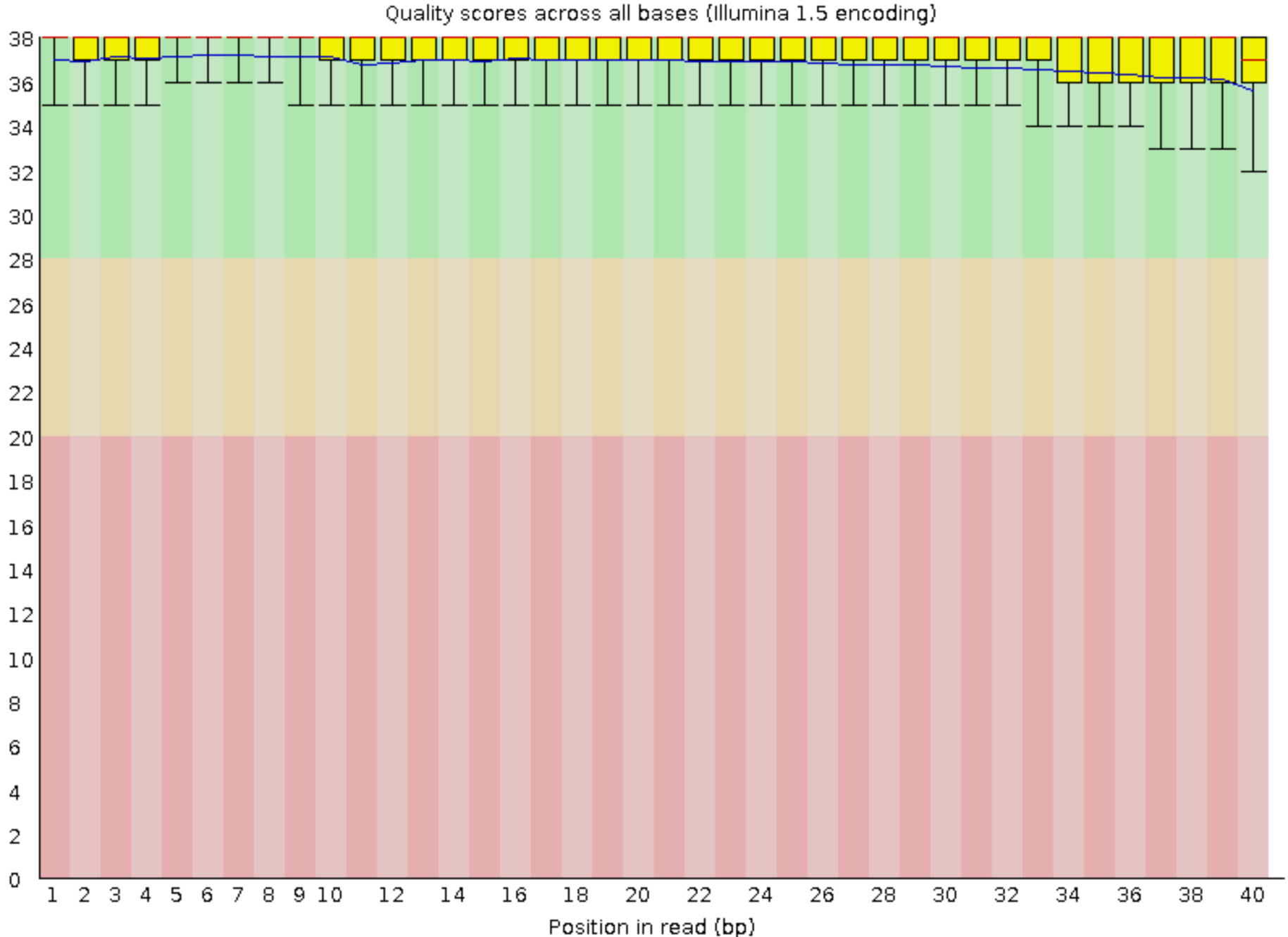
FastQC Report

Summary

- Basic Statistics
- Per base sequence quality
- Per sequence quality scores
- Per base sequence content
- Per base GC content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Kmer Content



Pré-processamento dos dados : Análise de qualidade



Pré-processamento dos dados



Poda de qualidade

Trimmomatic

http://www.usadellab.org/cms/?page=trimmomatic

Estatísticas de qualidade das leituras, poda de qualidade ou de adaptadores etc.

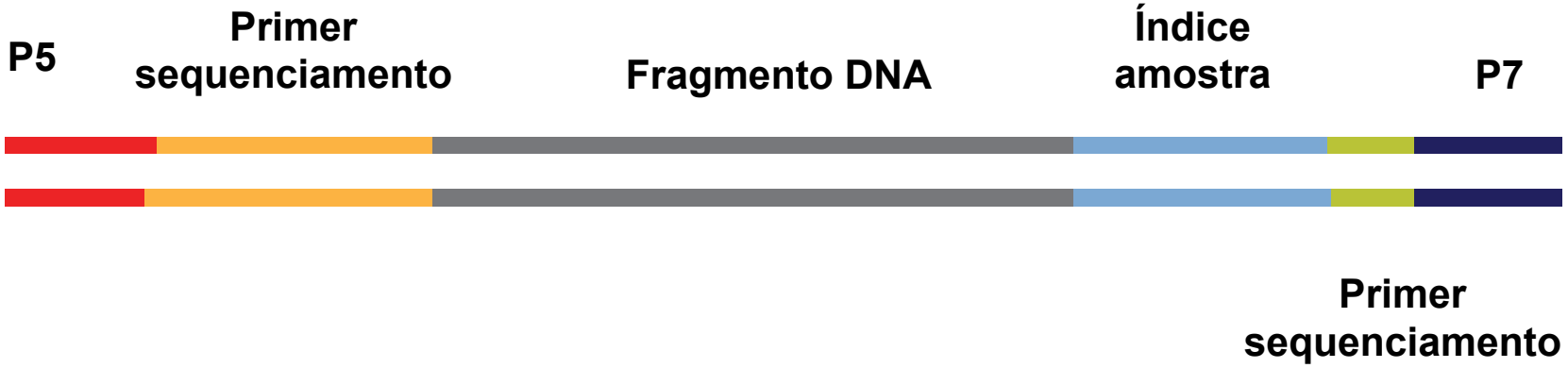
SeqyClean

https://bitbucket.org/izhbannikov/seqyclean

Poda de qualidade, de adaptadores, contaminantes…

FASTX-Toolkit, TrimGalore!, etc, etc

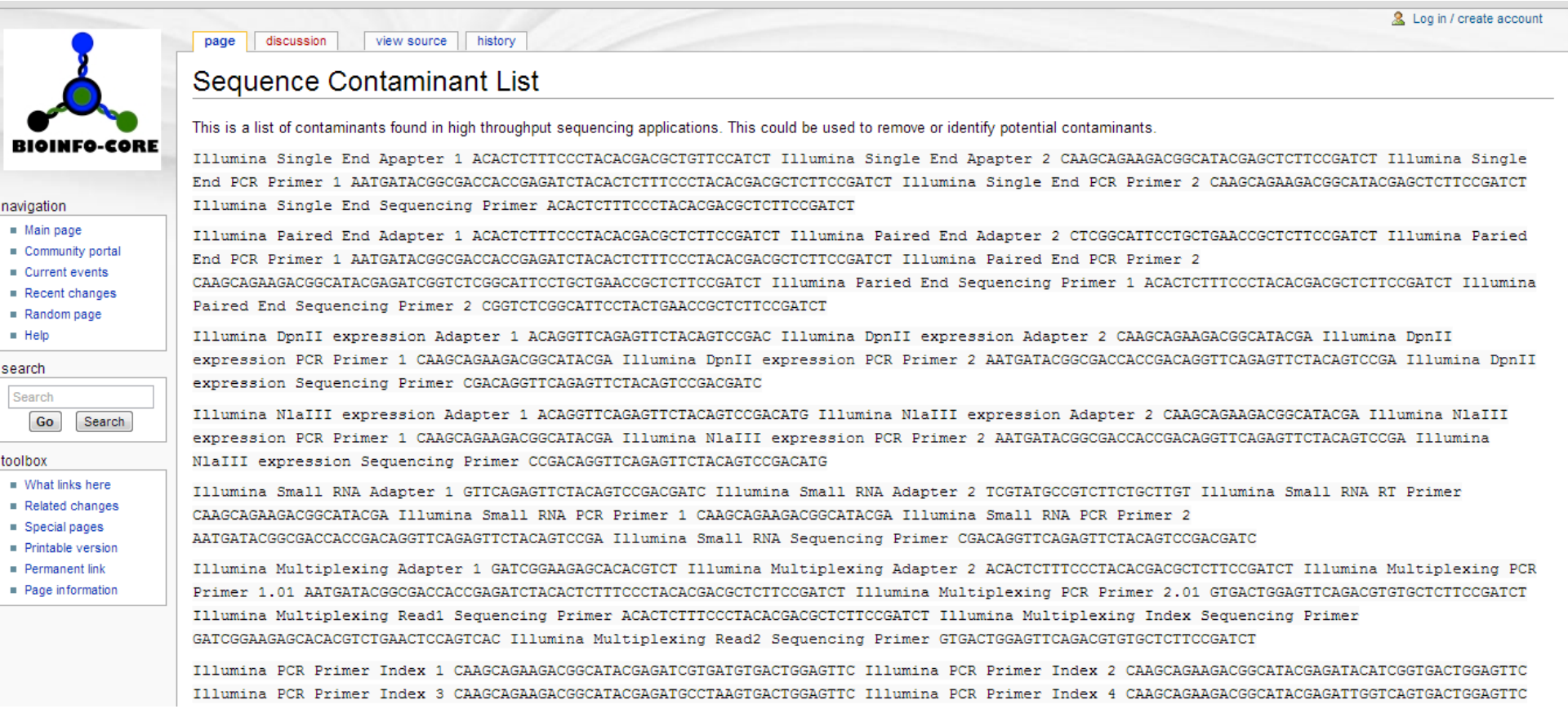
Poda dos adaptadores e índices



Pré-processamento dos dados

Poda dos adaptadores e índices

Lista dos adaptores etc. :
http://bioinfo-core.org/index.php/Sequence_Contaminant_List



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BIOINFO-CORE

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Sequence Contaminant List

This is a list of contaminants found in high throughput sequencing applications. This could be used to remove or identify potential contaminants.

Illumina Single End Apapter 1 ACACTCTTTCCCTACACGACGCTGTTCCATCT Illumina Single End Apapter 2 CAAGCAGAAGACGGCATACGAGCTCTTCCGATCT Illumina Single End PCR Primer 1 AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Single End PCR Primer 2 CAAGCAGAAGACGGCATACGAGCTCTTCCGATCT Illumina Single End Sequencing Primer ACACTCTTTCCCTACACGACGCTCTTCCGATCT

Illumina Paired End Adapter 1 ACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Paired End Adapter 2 CTCGGCATTCCTGCTGAACCGCTCTTCCGATCT Illumina Paried End PCR Primer 1 AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Paired End PCR Primer 2 CAAGCAGAAGACGGCATACGAGATCGGTCTCGGCATTCCTGCTGAACCGCTCTTCCGATCT Illumina Paried End Sequencing Primer 1 ACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Paired End Sequencing Primer 2 CGGTCTCGGCATTCCTACTGAACCGCTCTTCCGATCT

Illumina DpnII expression Adapter 1 ACAGGTTCAGAGTTCTACAGTCCGAC Illumina DpnII expression Adapter 2 CAAGCAGAAGACGGCATACGA Illumina DpnII expression PCR Primer 1 CAAGCAGAAGACGGCATACGA Illumina DpnII expression PCR Primer 2 AATGATACGGCGACCACCGACAGGTTCAGAGTTCTACAGTCCGA Illumina DpnII expression Sequencing Primer CGACAGGTTCAGAGTTCTACAGTCCGACGATC

Illumina NlaIII expression Adapter 1 ACAGGTTCAGAGTTCTACAGTCCGACATG Illumina NlaIII expression Adapter 2 CAAGCAGAAGACGGCATACGA Illumina NlaIII expression PCR Primer 1 CAAGCAGAAGACGGCATACGA Illumina NlaIII expression PCR Primer 2 AATGATACGGCGACCACCGACAGGTTCAGAGTTCTACAGTCCGA Illumina NlaIII expression Sequencing Primer CCGACAGGTTCAGAGTTCTACAGTCCGACATG

Illumina Small RNA Adapter 1 GTTCAGAGTTCTACAGTCCGACGATC Illumina Small RNA Adapter 2 TCGTATGCCGTCTTCTGCTTGT Illumina Small RNA RT Primer CAAGCAGAAGACGGCATACGA Illumina Small RNA PCR Primer 1 CAAGCAGAAGACGGCATACGA Illumina Small RNA PCR Primer 2 AATGATACGGCGACCACCGACAGGTTCAGAGTTCTACAGTCCGA Illumina Small RNA Sequencing Primer CGACAGGTTCAGAGTTCTACAGTCCGACGATC

Illumina Multiplexing Adapter 1 GATCGGAAGAGCACACGTCT Illumina Multiplexing Adapter 2 ACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Multiplexing PCR Primer 1.01 AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Multiplexing PCR Primer 2.01 GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT Illumina Multiplexing Read1 Sequencing Primer ACACTCTTTCCCTACACGACGCTCTTCCGATCT Illumina Multiplexing Index Sequencing Primer GATCGGAAGAGCACACGTCTGAACTCCAGTCAC Illumina Multiplexing Read2 Sequencing Primer GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT

Illumina PCR Primer Index 1 CAAGCAGAAGACGGCATACGAGATCGTGATGTGACTGGAGTTC Illumina PCR Primer Index 2 CAAGCAGAAGACGGCATACGAGATACATCGGTGACTGGAGTTC Illumina PCR Primer Index 3 CAAGCAGAAGACGGCATACGAGATGCCTAAGTGACTGGAGTTC Illumina PCR Primer Index 4 CAAGCAGAAGACGGCATACGAGATTGGTCAGTGACTGGAGTTC

Pré-processamento dos dados

